Reinforcement Learning for Simple Adaptive Robot Behavior

Literature review and project proposal

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Roadmap

Background

Why these papers

Paper 1: Gonzalez-Santocildes et al. (2024)

Paper 2: Zedda et al. ($2023 \rightarrow 2025$)

Paper 3: Wang et al. (2025)

Paper 4: Tozadore et al. (2018)

Synthesis

Platform and feasibility

Project directions

Questions for the committee

Background

Social robots and adaptive behavior

Robot: an embodied machine that senses, computes, and acts.

Social robot: interacts with people using human-like cues and social rules.

Human–Robot Interaction: the field — robotics, AI, psychology, design — studying how people and robots interact.

Core challenges

- Embodiment — the body shapes interaction
- Autonomy — how much the robot decides
- Trust — whether people rely on it

Signals we can use

- Speech, gesture, gaze, proxemics (space)
- Gaze, repositioning, touch — usable as a *reward*

The idea in one sentence

A robot can learn online, from a person's social signals as a reward, to adapt simple social behaviors — formalized as a Markov Decision Process and solved with lightweight RL.

Three ways a machine can learn

Supervised

- Learn input $\rightarrow$ correct label from examples.
- Needs large pre-labeled data; trained offline.
- E.g. an emotion classifier from labeled images.

Unsupervised

- Find structure in unlabeled data.
- Clustering, dimensionality reduction.
- E.g. group users by interaction style.

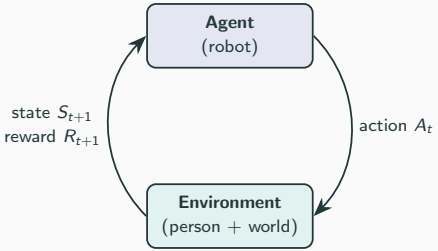
Reinforcement

- Learn by trial and error from a reward.
- No labels; adapts online.
- E.g. adjust distance if the person leans in or backs away.

Why reinforcement learning here

The "correct" social behavior is person-specific, unknown in advance, and shifts during interaction. Supervised learning needs a labeled answer up front and cannot adapt mid-interaction. RL learns the answer from the interaction itself.

The reinforcement-learning loop



At each step t the agent:

1. observes state S_t,
2. takes action A_t via policy $\pi(a \mid s)$,
3. receives reward R_{t+1},
4. lands in state S_{t+1}.

Goal: find π maximizing the expected discounted return

$$G_t = \sum_{k \geq 0} \gamma^k R_{t+k+1}, \quad \gamma \in [0, 1).$$

Value $Q(s, a)$ = expected return from taking a in s.

Formalized as a Markov Decision Process $\langle S, A, P, R, \gamma \rangle$: the next state depends only on the current (s, a). [Sutton2018]

Q-learning

Idea: learn the value of each (state, action) pair directly — no model of the world needed.

Q-learning update (Watkins & Dayan, 1992)

$$Q(s,a) \leftarrow Q(s,a) + \alpha\Big[\underbrace{r + \gamma \max_{a'} Q(s',a')}_{\text{target}} - Q(s,a)\Big]$$

α: learning rate γ: discount r: observed reward

Exploration vs. exploitation

- Must try new actions and use known-good ones.
- ε-greedy: random action with probability ε, else $\arg\max_a Q(s,a)$; decay ε over time.

Tabular vs. approximation

- Tabular: one value per (s,a) — good for small, discrete spaces.
- Large or continuous spaces $\rightarrow$ function approximation (Deep Q-Networks).

Policy gradients and human feedback

Policy gradients (REINFORCE)

- Directly tune a parameterized policy π_θ toward higher reward.
- Suited to continuous actions: distance, speed, timing.
- Used by Mitsunaga et al. to adapt proxemics, gaze, and timing on a social robot. [Mitsunaga2005]

Interactive RL (human feedback)

- The human supplies the reward signal.
- TAMER: non-experts shape an agent via approval / disapproval. [Knox2009]
- Cuts the number of trials needed — important on hardware.

The hardware reality

Every real trial costs time, battery, and joint wear. Sample efficiency matters, and policies that work in simulation can fail on the physical robot — the "reality gap".

Learning from demonstration

Teach by showing, not by coding or rewarding. [Argall2009]

- Intuitive for non-experts — people teach by demonstrating.
- Kinesthetic teaching: set joint stiffness to zero, physically move the robot's limb, record the trajectory, replay it.

Pairing it with RL

A demonstration gives a good starting policy; RL then refines it through interaction — easing RL's slow cold-start.

Personalization in long-term HRI

Why adapt at all?

- Short interactions hide the hard part: sustaining engagement over many sessions. [Irfan2019]
- Adaptation counters the novelty effect — interest fades once a robot stops being new.
- Built on user modeling and incremental learning. [Irfan2021]

Static vs. dynamic

- Static: fix a profile once (a questionnaire).
- Dynamic: adjust in real time to live social signals.

The ethical edge [Pollmann2023]

The same personalization that improves user experience can be used to steer a person against their own autonomy. Adaptive systems need explicit ethical guardrails.

A recurring pattern: games and simulation

As we read across this field, two patterns showed up again and again — worth naming before the papers:

- **Games as metaphor and as tooling.** Game engines and "serious games" recur everywhere — Unity (Gonzalez), *Assistive Gym* (Wang), a cooking game (Zedda), a storytelling activity (Tozadore).
- **It cuts across robot types.** The same toolkit serves social humanoids (NAO, Pepper) *and* industrial cobots (the UR3e arm) alike.
- **Simulation is nearly unavoidable** — it is safe, cheap, and fast for the many trials RL needs.

The catch

Training in simulation always introduces the *reality gap* (sim-to-real): a policy that works in a clean simulator can fail on a noisy physical robot. Every one of these papers has to confront it — and so will we.

Why these papers

Why these papers

Foundational references (some older)

- Used to build the vocabulary in the previous slides: RL, learning from demonstration, personalization, ethics.
- Deliberately allowed to be older — core definitions and methods do not expire.

Four modern papers we analyze in depth

- Tozadore (2018): adaptation on a humanoid (NAO), rule-based — the pre-RL baseline.
- Gonzalez-Santocildes (2024): RL learns comfort-based adaptation.
- Zedda (2023→2025): RL for engagement on a humanoid, simulation *then* a real 28-user study — closest task match.
- Wang (2025): preference-based RL — the current frontier.

Why this set

Together they trace the field's trajectory — rule-based → RL → preference-based; hand-designed → learned reward; simulation → real study — and let us locate where our own project sits.

Paper 1: Gonzalez-Santocildes et al. (2024)

Problem and setup

Adaptive Robot Behavior Based on Human Comfort Using Reinforcement Learning, IEEE Access. [Gonzalez2024]

The problem

- Most RL for robotics ignores the human — it optimizes task success, not comfort.
- Prior comfort work is largely theoretical and not replicable.

Contribution

- An open-source, parametric Unity environment the community can reuse.
- A comfort reward function whose strictness is set by one parameter.

Setup (worth noting)

- Robot: UR3e — an industrial arm, not a social humanoid.
- Task: pick-and-place near a moving simulated human.
- Comfort here is physical (distance, speed), not social.
- Algorithm: SAC (deep RL for continuous control).

Methodology

Standard task reward, minus a comfort penalty the user can switch on and weight:

$$R = k_1 F_{\text{task1}} + k_2 F_{\text{task2}} + k_3 F_{\text{task3}} \; - \; F_{\text{comfort}}(C_l, N_c)$$

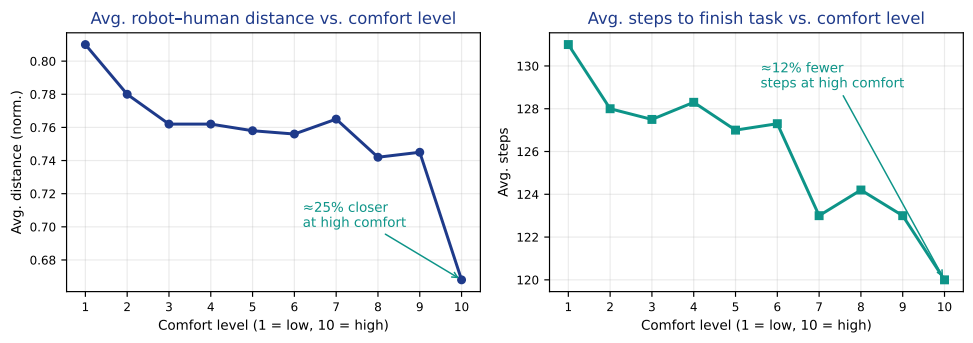
$$F_{\text{comfort}}(C_l, N_c) = \alpha \, F_{\text{conf1}}(C_l, N_c) \; + \; \beta \, F_{\text{conf2}}(C_l, N_c)$$

- F_{conf1}: penalty on distance (too close hurts reward).
- F_{conf2}: penalty on joint speed (too fast hurts reward).
- α, β: weights; set both to 0 and it is plain task RL.

Why "comfort levels" (C_l of N_c)?

- Comfort is discretized (10 levels), each trained separately — easier than one policy over a continuum.
- The level sets where the penalty threshold sits.
- Static per run = controlled variation: change one thing, attribute the effect cleanly.

Results



As comfort rises, the robot works closer (≈25% less distance) and faster (≈12% fewer steps). Trained with SAC (30M steps, ≈4.4 days on a GPU). *Charts redrawn from Gonzalez et al. (2024), Figs. 5–6; values approximate.* [Gonzalez2024]

Critique and relevance

Strengths

- Clean controlled result: behavior tracks the comfort parameter.
- Open, replicable environment.

Gaps

- Comfort is set manually; measuring it live from a person is unsolved (they suggest HR/EEG/a button).
- Only move-pick-place tasks.
- Physical/industrial comfort, not social; not a humanoid.
- Validated only in simulation.

Relevance to us

- We borrow the proven idea — comfort in the reward — and add the missing piece: a live comfort signal from a robot's own sensors.
- We move from physical toward social comfort.
- A small discrete state space lets us use tabular Q-learning rather than SAC: lighter and laptop-trainable.

Paper 2: Zedda et al. (2023 $\rightarrow$ 2025)

Problem and setup

An Adaptive Behaviour-Based Strategy for SARs interacting with Older Adults with MCI..., SARs workshop @ CHI 2023 [Zedda2023] — extended as *Adaptive Humanoid Robot Behaviour... through Reinforcement Learning*, *Behaviour & Information Technology*, 2025. [Zedda2025]

The problem

- Cognitive training for older adults with MCI is repetitive — leading to disengagement and dropout.
- A socially assistive robot can help — *if* it adapts its behavior to the user's state.

Contribution

- A Q-learning strategy that picks verbal and non-verbal actions to keep engagement high.
- Two selectable robot personalities (extravert / introvert).

We follow one team across two papers

- **2023 (workshop):** the method, validated in *simulation only* — real study deferred.
- **2025 (journal):** same method, now run as a real **28-user** study — the gap the workshop left open.
- Robot: Pepper — NAO's sibling (same maker, same NAOqi SDK), so methods transfer.
- Task: a "cooking" serious game (8 questions in 2023, 16 in 2025).

Methodology — the RL model

Tabular Q-learning, $Q(s, a)$ table, ε-greedy; model-free, off-policy. State = engagement $\times$ answer $\Rightarrow$ 6 states; 3 actions.

State (6)

- Engagement L / I / H, derived from gaze + smile
- Answer right / wrong
- $\Rightarrow$ six (engagement, answer) states

Actions (3)

- engaging / dynamic
- neutral
- stimulating (re-engage)

each filtered through personality: pitch, rate, volume, gesture

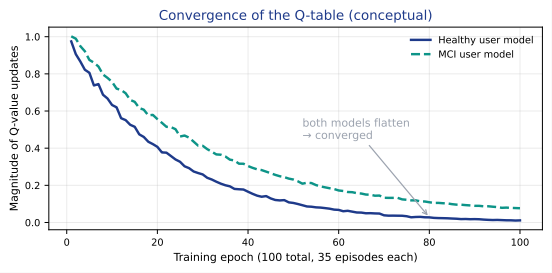
Reward $R(s, a)$

- Engagement: $H{=}+10$, $I{=}-2.5$, $L{=}-5$
- Answer: right $+10$, wrong -5
- weighted by user-profile probabilities; $-0.05t$ per step

A detail we tracked across both versions

The discount rose from $\gamma = 0.05$ (2023) to $\gamma = 0.2$ (2025) — still myopic (the agent mostly values the *immediate* engagement response, close to a contextual bandit), but less extreme. Learning rate stays high at $\alpha = 0.8$; ε decays from 1 to 0.2.

Results — simulation (the convergence check)



Training: Python; 100 simulated users × 35 iterations; two user models (healthy, MCI) plus "optimal" and "inactive" probe profiles. **Outcome**

- Converges within ∼15 iterations; the QL policy beats the random policy for *every* profile.
- MCI plateaus below healthy — expected, as MCI users disengage more easily.
- Learned Q-table: MCI users need more *stimulating* actions, healthy users more *engaging* ones.

Curves redrawn from Zedda et al. (2025), Fig. 3 (four user profiles; QL solid vs. random dashed). [Zedda2025]

Results — the real 28-user study (new in 2025)

Within-subjects: each of 28 users (age 20–72) met both an **adaptive** robot (trained Q-table) and a **random** robot; order and personality counterbalanced; ethics-approved.

RQ1 — can users tell?

- Yes. Adaptive rated more adaptive in animations ($p \approx 0.0001$) and verbal feedback ($p \approx 0.0005$).

RQ2 — engagement / UX

- **86%** preferred the adaptive robot.
- Higher engagement (UES) and enjoyment (4.57 vs 3.96).
- Comfort difference *not* significant.

RQ3 — perception (Godspeed)

- Likeability 4.50 vs 3.94 *
- Anthropomorphism 3.6 vs 2.90 *
- Perceived intelligence 3.89 vs 3.6 (n.s.)

Simulation showed the algorithm *converges*; this study shows the interaction *works* — users notice, prefer, and warm to the adaptive robot. ($*$ significant, $p < 0.05$.) [Zedda2025]

Critique and relevance

Strengths

- The closest match to our project: humanoid + tabular Q-learning + engagement reward — now carried all the way to a *real* study.
- Clean simulation-then-real pipeline; reusable instruments (within-subjects design, UES, Godspeed).

Gaps

- Reward weights still hand-set from prior qualitative work (their own stated future work).
- Only 3 actions; personality fixed per session; low γ limits long-horizon credit.
- Single short session, N=28, lab setting, no formal cognitive assessment.

Relevance to us

- Validates our method *end-to-end*: tabular Q-learning for engagement on a humanoid works with real users, not just in simulation.
- Pepper and NAO share the NAOqi SDK, so their state and actions map across.
- Gives us a proven study template: adaptive vs. random, within-subjects, with UES + Godspeed.

Paper 3: Wang et al. (2025)

Robots in Today's World

- Robots are widely used:
  - Delivery
  - Assistance
  - Companionship
- Machine learning improves robot intelligence
- But human interaction remains challenging

Industrial vs Human-Robot Systems

Industrial Robots

- Fixed tasks
- Controlled environments
- Repetitive behavior

Human-Interactive Robots

- Dynamic environments
- Must adapt to users
- No single correct behavior

Why Personalization Matters

Same task, different users

- Feeding a baby $\rightarrow$ fast movement
- Feeding an elderly person $\rightarrow$ slow and careful

Key insight: One behavior does not fit all users

Rule-Based Methods

Early approach

- Hand-crafted rules
- Example: spoon $\rightarrow$ mouth movement
- Food $\rightarrow$ pick and serve

Limitations

- Cannot cover all situations
- Complex real-world interactions
- High design cost

Learning-Based Methods

- Robots learn from experience
- Improve over time
- Based on Reinforcement Learning

Preference-Based Reinforcement Learning

PbRL idea

- Show two behaviors to user
- User selects preferred one
- Robot learns preferences

Advantage: No manual reward design

Problem: Requires learning from scratch

Problem with Fine-Tuning

Issue

- Pretrained robots already know tasks

But fine-tuning causes:

- Loss of learned skills
- Performance degradation

Example: Safe drinking becomes unstable after adaptation

Key Trade-off

- Training from scratch $\rightarrow$ too much data
- Fine-tuning $\rightarrow$ damages performance

Need: A better balance between both

PbARL Idea

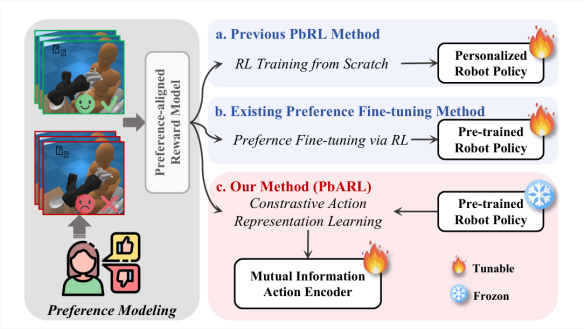
Core concept

- Keep pretrained policy fixed
- Add adaptation layer

Key idea:

- Learn action representations
- Adjust behavior via preferences

Result: Personalization without forgetting skills



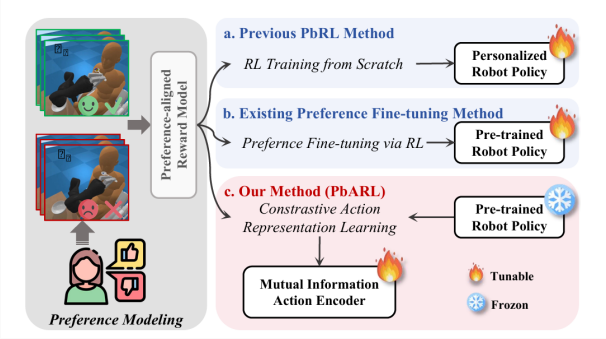
Latent Action Space

- Map actions into latent space

Captures:

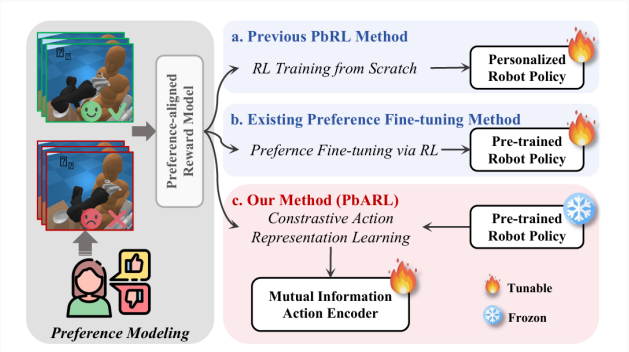
- Fast ↔ slow
- Safe ↔ risky
- Gentle ↔ aggressive

Benefit: Easier preference modeling



PbARL Components

1. Pretrained policy (task knowledge)
2. Preference reward model
3. Conditional VAE (cVAE)



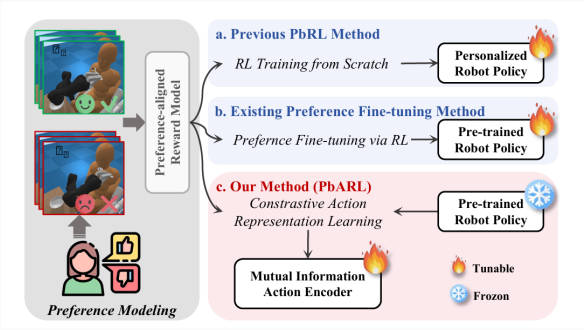
Conditional VAE (cVAE)

- Encoder: (state, action) $\rightarrow$ latent space
- Decoder: latent $\rightarrow$ reconstructed action

Meaning:

- Understand actions
- Reconstruct actions

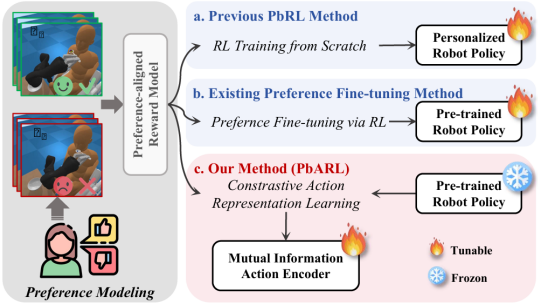
Benefit: State-aware and realistic behavior



Training Objectives

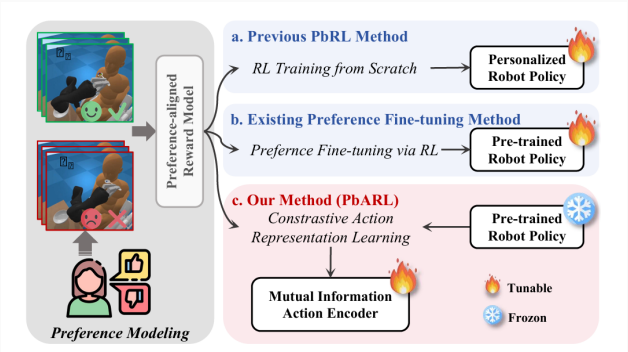
- Reconstruction loss → preserve actions
- Preference loss → match user preferences
- KL loss → smooth latent space
- Dynamics loss → physical feasibility

Goal: Safe and personalized behavior



PbARL Pipeline

- Observe state
- Generate candidate actions
- Encode into latent space
- Evaluate preferences
- Re-rank actions
- Select best action



Experiments Setup

Tasks:

- Feeding
- Drinking
- Scratching

Users:

- Neutral
- Cautious
- Impatient

Methods:

- Pretrained policy
- PbRL-HRI
- Fine-tuning
- PbARL

Results

- PbRL-HRI → needs many feedback samples
- Fine-tuning → reduces performance
- Pretrained → not personalized
- PbARL → best trade-off

Conclusion: Best balance between performance and personalization

TABLE I

PERFORMANCE OF EACH MODEL ACROSS TASKS AND USER TYPES IN TERMS OF SUCCESS RATE (SUCC%) AND MEAN PREFERENCE REWARD RETURNS (REW) ON 500 UNSEEN TEST SCENARIOS. THE CHANGES COMPARED TO THE PRE-TRAINED MODEL ARE SHOWN IN THE Δ COLUMNS WITH ARROWS (↑ FOR IMPROVEMENT, ↓ FOR DECREASE). BEST RESULTS (EXCLUDING PRE-TRAINED) ARE BOLDED.

		Neutral User				Cautious User				Impatient User			
Task	Method	Succ	Δ	Rew	Δ	Succ	Δ	Rew	Δ	Succ	Δ	Rew	Δ
Feeding	Pre-trained	92	–	11.2	–	90	–	12.3	–	89	–	15.6	–
	PbRL-HRI	41	↓51	12.1	↑0.9	42	↓48	12.8	↑0.5	39	↓50	18.9	↑3.3
	PrefFT	73	↓19	45.6	↑34.4	69	↓21	51.2	↑38.9	75	↓14	44.8	↑29.2
	PbARL(n=1)	80	↓12	44.8	↑33.6	79	↓11	48.1	↑35.8	78	↓11	46.2	↑30.6
	PbARL(n=20)	84	↓8	**53.2**	**↑42.0**	83	↓7	52.1	↑39.8	**85**	**↓4**	55.7	↑40.1
	PbARL	**88**	**↓4**	50.7	↑39.5	**83**	**↓7**	**54.7**	**↑42.4**	84	↓5	**56.1**	**↑40.5**
Drinking	Pre-trained	79	–	14.6	–	78	–	16.1	–	81	–	14.2	–
	PbRL-HRI	29	↓50	16.3	↑1.7	35	↓43	17.5	↑1.4	30	↓51	14.7	↑0.5
	PrefFT	63	↓16	41.2	↑26.6	61	↓17	40.7	↑24.6	70	↓11	36.9	↑22.7
	PbARL	**74**	**↓5**	**51.8**	**↑37.2**	**70**	**↓8**	**48.6**	**↑32.5**	**79**	**↓2**	**51.1**	**↑36.9**
Scratching	Pre-trained	71	–	10.9	–	74	–	11.8	–	74	–	10.2	–
	PbRL-HRI	23	↓48	13.2	↑2.3	21	↓53	14.5	↑2.7	24	↓50	13.9	↑3.7
	PrefFT	55	↓16	**43.3**	↑32.4	56	↓18	46.3	↑34.5	52	↓22	47.5	↑37.3
	PbARL	**67**	**↓4**	40.6	↑29.7	**69**	**↓5**	**51.6**	**↑39.8**	**70**	**↓4**	**51.1**	**↑40.9**

Ablation Study

- Single action → poor performance
- Multiple candidates → better results

Insight: More options improve preference matching

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Real Robot Experiments

Setup:

- Kinova Jaco arm
- RealSense camera
- Mouth tracking system
- Safety stop mechanism

Results:

- Higher user satisfaction
- Better personalization
- Stable behavior

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	PbRL-HRI	41	↓51	12.1	↑0.9	42	↓48	12.8	↑0.5	39	↓50	18.9	↑3.3
	PrefFT	73	↓19	45.6	↑34.4	69	↓21	51.2	↑38.9	75	↓14	44.8	↑29.2
	PbARL(n=1)	80	↓12	44.8	↑33.6	79	↓11	48.1	↑35.8	78	↓11	46.2	↑30.6
	PbARL(n=20)	84	↓8	**53.2**	**↑42.0**	83	↓7	52.1	↑39.8	**85**	**↓4**	55.7	↑40.1
	PbARL	**88**	**↓4**	50.7	↑39.5	**83**	**↓7**	**54.7**	**↑42.4**	84	↓5	**56.1**	**↑40.5**
Drinking	Pre-trained	79	–	14.6	–	78	–	16.1	–	81	–	14.2	–
	PbRL-HRI	29	↓50	16.3	↑1.7	35	↓43	17.5	↑1.4	30	↓51	14.7	↑0.5
	PrefFT	63	↓16	41.2	↑26.6	61	↓17	40.7	↑24.6	70	↓11	36.9	↑22.7
	PbARL	**74**	**↓5**	**51.8**	**↑37.2**	**70**	**↓8**	**48.6**	**↑32.5**	**79**	**↓2**	**51.1**	**↑36.9**
Scratching	Pre-trained	71	–	10.9	–	74	–	11.8	–	74	–	10.2	–
	PbRL-HRI	23	↓48	13.2	↑2.3	21	↓53	14.5	↑2.7	24	↓50	13.9	↑3.7
	PrefFT	55	↓16	**43.3**	↑32.4	56	↓18	46.3	↑34.5	52	↓22	47.5	↑37.3
	PbARL	**67**	**↓4**	40.6	↑29.7	**69**	**↓5**	**51.6**	**↑39.8**	**70**	**↓4**	**51.1**	**↑40.9**

Conclusion

Key idea

- Do not change the robot
- Change action representation

Final takeaway: Personalization without forgetting

Paper 4: Tozadore et al. (2018)

Problem and Setup

Towards Adaptation and Personalization in Task-Based Human-Robot Interaction, LARS/SBR 2018. [Tozadore2018]

The problem

- Social robots lose novelty over time.
- Users become less engaged and less motivated.
- Question: can personalization sustain engagement?

Contribution

- Dynamic adaptation system for HRI.
- Continuously adjusts robot behavior using user signals.
- Builds user-specific interaction history.

Setup (worth noting)

- Robot: NAO humanoid robot.
- Task: storytelling interaction.
- 10 participants (12 min sessions).
- Rule-based decision system (no RL).

Motivation

Why adaptation matters

- Users differ in preferences and engagement levels.
- One fixed robot behavior is insufficient.
- Engagement decreases without personalization.

Analogy

- Netflix $\rightarrow$ personalized recommendations
- YouTube $\rightarrow$ adaptive content feed
- Learning systems $\rightarrow$ adaptive difficulty

Goal in robotics

- Detect user state in real time.
- Adapt interaction style dynamically.
- Maintain long-term engagement.

System Overview

Main idea

- The robot continuously monitors:
  - Attention
  - Communication
  - Learning performance
- Then adapts behavior accordingly.

Feedback loop

- Observe user → Estimate state → Compute score → Adapt behavior

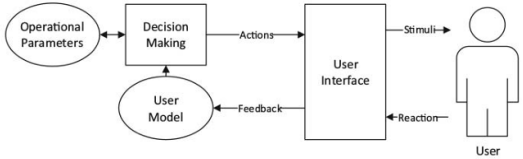


Figure 1. Adaptive system scheme according to [2].

User State Estimation

Attention (α)

- Face gaze tracking
- Posture analysis
- Absence from camera

Communication (β)

- Number of words
- Emotion recognition
- Facial expression classification

Learning (γ)

- Answer correctness
- Response time
- Task success rate

Table I
OBJECTIVE MEASURES BY GROUP.

Attention (α)	**Communication (β)**	**Task (γ)**
Face gaze (Fg)	Number of Words (nW)	Success Rate (Sr)
Posture (P)	Emotions (Em)	Time to answer (Tta)

Emotion Recognition Module

Model

- MobileNet (lightweight CNN)
- Trained on FER+, RAF, QIDER datasets
- Real-time emotion classification

Emotion classes

- Happy, Sad, Angry, Fear
- Surprise, Disgust, Neutral

Why MobileNet?

- Fast inference
- Low computational cost
- Suitable for robotics hardware

Face Gaze and Answer Evaluation

Face gaze tracking

- Haar cascade face detection
- Measures time away from robot
- Long absence $\rightarrow$ reduced attention

Answer evaluation

- Levenshtein distance
- Measures text similarity
- Small distance $\rightarrow$ correct answer

Adaptation Function

Core equation

$$F_{Adp}(t) = w_\alpha \alpha(t) + w_\beta \beta(t) + w_\gamma \gamma(t)$$

- α: attention
- β: communication
- γ: learning performance
- Weights: 0.2, 0.2, 0.6

Decision rule

- $\geq 0.66 \rightarrow$ increase interaction
- $\leq 0.33 \rightarrow$ decrease interaction
- otherwise $\rightarrow$ maintain behavior

Robot Behavior States

- State 1: minimal interaction, low gestures
- State 2: slightly interactive
- State 3: neutral behavior
- State 4: more expressive
- State 5: highly social and engaging

Goal: dynamically shift between states based on user engagement

Experiment Setup

Setup

- NAO humanoid robot
- 10 participants (students)
- 12-minute interactions

Task

- Storytelling activity
- Questions after each story

Collected signals

- Attention
- Communication
- Learning performance

Results and Findings

- Users reported high enjoyment.
- Perceived personalization improved engagement.
- Adaptive behavior improved interaction quality.

Key insight: Personalization increases perceived social presence of the robot.

Limitations and Future Work

Limitations

- Small sample size (10 users)
- Short interaction time
- Emotion confusion (concentration vs sadness)
- Weak semantic understanding

Future work

- Better emotion recognition models
- Long-term studies
- Improved language understanding
- Automatic user identification

Takeaway

Core message

- Observe user behavior
- Estimate engagement (attention, communication, learning)
- Adapt robot behavior dynamically
- Personalize interaction using user memory

Final idea: Adaptive + personalized robots can sustain human engagement over time.

Synthesis

Gaps across the four papers

- **Reward design:** hand-designed (Tozadore, Gonzalez, Zedda) is brittle; learned preferences (Wang) are powerful but heavy. *We start with a simple explicit reward; learned preferences are future work.*
- **Simulation vs. real:** simulation dominates training (Gonzalez, Wang's benchmark); real studies are still small (Tozadore n=10, Wang N=8, Zedda N=28). *We go simulation-first, then a small real study.*
- **Platform:** industrial arms (Gonzalez, Wang) vs. humanoids (Zedda's Pepper, Tozadore's NAO). *We target a humanoid.*
- **Method complexity:** deep RL on GPUs (Gonzalez, Wang) vs. lightweight tabular (Zedda). *We stay interpretable and feasible.*
- **Evaluation:** subjective surveys dominate. *We combine objective sensor metrics with subjective scales.*

Our project sits where these gaps overlap: a humanoid, a simple interpretable RL method, a live reward signal, and an evaluation that triangulates both kinds of metric.

Platform and feasibility

Platform and feasibility

We have kept our setup robot-agnostic, because a few practical questions are still open and the answers shape the project:

- Is **NAO** still the platform you would like us to use in 2026, or is there a newer or preferred one?
- Will a physical NAO realistically be **available before the end of the semester**?
- If not, should we focus **entirely on simulation**?
- And should simulation stay **NAO-compatible**, or are entirely different directions acceptable?

Our initial setup does not assume NAO specifically, so we can adapt to whichever answer you give.

What NAO offers, if it is the platform

Hardware [Aldebaran]

- 58 cm, ∼5.6 kg, 25 degrees of freedom
- Dual cameras; 4 microphones
- Sonar distance sensors (∼25–255 cm)
- Tactile sensors (head + hands); foot force sensors; IMU
- Battery ∼45–90 min; limited onboard compute

Software (NAOqi SDK)

- `ALMotion` — joints, posture, walking
- `ALMemory` / `ALSensors` — sensor access
- `ALSonar` — distance; `ALTextToSpeech` — speech
- Choregraphe IDE (with a virtual NAO)
- Python / C++ SDKs; ROS support

Sonar and tactile sensors give a clean, real-time signal for a comfort- or engagement-based reward — which is what makes a humanoid like NAO a natural fit *if* it is available.

Project directions

Candidate directions

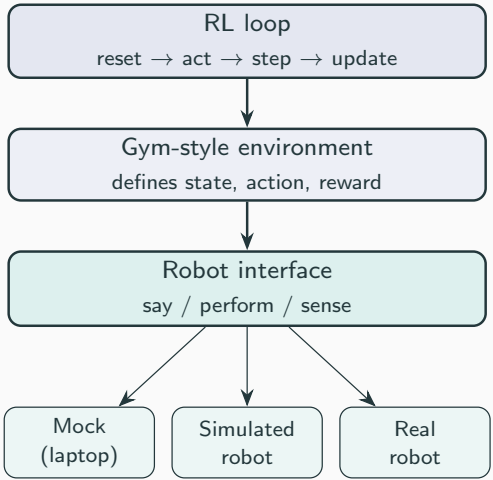
We are not committing yet — these share one RL loop; only the state, actions, and reward differ. We would like the discussion to help pick one (or suggest another).

Direction	Adaptive behavior	Reward signal	Baseline
Greeting / social distance	Approach distance + greeting intensity	Tactile "approve" + sonar distance	Fixed scripted greeting
Tutoring / pace	Speech rate, pauses, difficulty	Answer correctness + engagement	Fixed-pace tutor
Engagement	Pick from a behavior repertoire	Change in gaze + proximity	Random / fixed rotation

Illustrated here on a NAO-like humanoid, but the same structure ports to a simulator or another platform.

Initial technical setup

We have started building a small, robot-agnostic RL loop. We expect most of it to be reusable across whichever direction we choose — it is early groundwork, not a finished system.



- The learning code talks to an abstract *interface*, not a specific robot.
- An early prototype runs in simulation against a mock that includes a simulated person.
- Sensor noise is injected to make the policy more robust.
- Moving to a simulated or real robot mainly means providing a new implementation of the interface.

A robot-agnostic interface (early prototype)

The interface is the seam that lets the same learning code run against a mock, a simulator, or a real robot. The mock includes a simulated person, so the loop runs with no hardware.

```
class RobotInterface(ABC):            # shared by mock / sim / real
    def perform(self, action): ...    # 'closer','back','intensity_up'...
    def sense(self) -> Signals: ...   # distance, touch, present

class MockRobot(RobotInterface):      # laptop + simulated person
    def sense(self):
        return Signals(distance=self._d + noise,
                       touch=self._person_approves(),
                       present=self._present)
```

Early result: in simulation it learns to approach to a comfortable distance and hold, rather than crowd the person — a clear improvement over a fixed script. Hosted on GitHub.

An illustrative MDP (greeting)

State (discretized)

- Distance bin: intimate / personal / social
- Current greeting intensity

Action

- move closer / move back
- increase / decrease intensity / hold

Reward

- $+$ head touch, sustained gaze, user stays
- $-$ user steps back, no touch, leaves

Algorithm

- Tabular Q-learning, ε-greedy
- Small discrete space, so it converges quickly

Evaluation

Study design

- Within-subject: each participant meets the scripted and the learned robot (counterbalanced).
- Higher power; controls individual differences.

Objective metrics

- Convergence speed
- Final interaction distance, approach latency
- Engagement duration

Subjective metrics

- Godspeed questionnaire (likeability, perceived safety, perceived intelligence)
- Comfort / trust survey

The point

- Triangulate: pair sensor metrics with subjective scales — addressing the field's main critique.

Implementation plan

1. **Mock / Gym environment** (platform-agnostic): encode the state–action–reward MDP and verify the agent learns.
2. **Simulated humanoid**: move to Choregraphe's virtual NAO or Webots — or another simulator if preferred.
3. **Real robot, if available**: fine-tune for a small number of trials; watch joint heat and battery.
4. **If NAO is not available**: stay in simulation, or target a different platform — the architecture does not assume NAO.

Code is on GitHub, with a modular interface so the same RL loop runs against mock, simulator, or hardware.

Questions for the committee

Questions for the committee

Platform (recap)

- NAO or another platform? Available this semester?
- Simulation-only acceptable if not?

Direction

- Which direction should we pursue — greeting, tutoring, engagement, or something you suggest?

Study logistics

- May we recruit fellow students as participants?
- What ethics / consent process is required?

Compute

- Tabular Q-learning runs on a laptop — no GPU needed.
- Only a deep-RL extension would need compute — is any available?

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A closing thought

If the robot's reward is our approval,

it learns what pleases us — not necessarily what is good for us.

Designing for that gap — comfort versus manipulation —

may be the real research question behind adaptive robots.

Thank you

Nour Gaser & Eman Saleh

Appendix

Held in reserve for discussion and deeper questions.

1. One Q-learning update, worked numerically
2. Supervised vs. RL on the same task
3. Why tabular Q-learning rather than deep RL
4. Preference-based RL in brief
5. Ethics: entertainment vs. manipulation
6. The reality gap

One Q-learning update, step by step

Setup (greeting MDP). $\alpha = 0.5$, $\gamma = 0.9$. Current estimate $Q(s_{\text{far}}, \texttt{closer}) = 0.0$.

1. **State** s_{far}: person in the social distance bin.
2. **Action** (via ε-greedy, exploiting): $a = \texttt{move closer}$.
3. **Outcome:** person stays and gives a head touch, so reward $r = +1$.
4. **Next state** $s' = s_{\text{personal}}$, where the current best value is $\max_{a'} Q(s', a') = 0.4$.

Apply the update

$$Q(s_{\text{far}}, \texttt{closer}) \leftarrow 0.0 + 0.5\big[1 + 0.9(0.4) - 0.0\big] = 0.5\,(1.36) = 0.68$$

The estimate rises from 0.0 to 0.68: "moving closer from far was good here." Repeat over episodes and the table converges to a policy that approaches each person's comfortable distance. [Sutton2018]

Supervised vs. RL on the same task

Task: choose how close the robot stands when greeting a new person.

	Supervised learning	**Reinforcement learning**
Data needed	A large labeled set (person features $\rightarrow$ "correct" distance), gathered in advance	None up front — learns from live reward (touch, sonar, dwell)
When it learns	Offline, before deployment	Online, during interaction
New person	Predicts from a fixed model; cannot adjust if wrong	Tries, observes feedback, corrects within the session
Failure mode	Confidently wrong on anyone unlike the training data	Some early exploration cost before convergence

The "right" social distance is personal and unknown in advance — the setting where RL's online adaptation fits better than a pre-trained classifier.

Why tabular Q-learning rather than deep RL

Our setting

- Small, discrete state and action space.
- A handful of distance bins and intensity levels.

So tabular fits

- One value per (state, action); updated with simple arithmetic.
- Runs on a laptop CPU; no GPU.
- Converges in seconds; fully interpretable (we can read the table).

When deep RL is needed

- Large or continuous spaces (Gonzalez, Wang) — hence SAC and GPUs.
- More expressive, but heavier, slower, and harder to interpret.

Starting tabular keeps the project feasible and explainable; deep RL is a clear future step if the state space grows.

Preference-based RL in brief

A likely question after Paper 3: *why not use preference-based RL?*

- **Idea:** instead of designing a reward, show the person two short behaviors and ask which they prefer. A model learns a reward from many such comparisons (Bradley–Terry).
- **Strength:** avoids hand-designing a reward — the hardest part in the other papers.
- **Cost:** needs many comparisons and, in practice, deep RL and real compute (Wang: 3 GPUs).

Our position

For a first project we use a simple explicit reward (touch, sonar, dwell) that is clean and interpretable. Replacing it with learned preferences is a natural extension once the basic loop works.

Ethics: entertainment vs. manipulation

The risk [Pollmann2023]

- Personalization that improves experience can also steer behavior against a person's autonomy.
- A robot optimizing "engagement" could learn manipulative tactics if the reward is naive.
- Vulnerable users (children, elderly) raise the stakes.

Our safeguards

- Reward tied to comfort, not raw attention.
- Transparency: the person knows the robot is adapting.
- Informed consent and a right to stop, for every participant.
- Store only what the task needs.

The reality gap

We prototype in simulation first, so we are clear about what that does and does not prove.

Why a gap exists

- Real sonar and tactile readings are noisy; mocks are clean.
- Real people behave unpredictably; scripted "users" do not.
- Joint dynamics, latency, and timing differ on hardware.

How we reduce it

- Inject noise into the mock sensors.
- Keep the state and action space small and discrete.
- Treat simulation as a convergence check; reserve real trials for fine-tuning.
- Report simulation and real results separately.

Simulation shows the algorithm works; only the physical robot shows the interaction works.